

A quick guide to FRESH MARKER

What is FreshMarker?

FRESH MARKER is a simple and embedded Java 21 template engine, which is inspired by *FreeMarker*. Most of the basic concepts are the same.

Integration

Maven Dependency

```
<dependency>
  <groupId>de.schegge</groupId>
  <artifactId>freshmarker</artifactId>
  <version>1.9.0</version>
</dependency>
```

Template Evaluation

```
String content = "Hello ${first} ${last}!";
Map<String, Object> model = Map.of(
    "first", "Jens", "last", "Kaiser");
Configuration configuration = new Configuration();
TemplateBuilder builder = configuration.builder();
Template template = builder.getTemplate("tst", content);
String output = template.process(model);
```

This template produces the output "Hello Jens Kaiser!".

Template Reduction

Templates can be partially evaluated. The newly created template saves you from having to evaluate static values again later.

```
Map<String, Object> model = Map.of("first", "Jens");
Template reduced = template.reduce(model);
```

This new template works internally on "Hello Jens \${last}!".

Model

The model contains all the data, that can be used in the template. A large number of types are automatically supported so that no conversions need to take place when using your own data.

Unless otherwise stated, the names refer to the corresponding classes in *JDK*.

Primitives

Primitives are all **Java** types whose value can be output directly in the template.

String, StringBuilder, StringBuffer, Boolean, AtomicBoolean, Byte, Short, Integer, Long, Float, Double, BigInteger, BigDecimal, AtomicInteger, AtomicLong, Locale, URI, URL, UUID, Date, Time, LocalDate, LocalTime, LocalDateTime, ZonedDateTime, Instant, Period, Duration, Year, YearMonth, MonthDay

Literals of Boolean, String, Integer and Double can be used in the template.

Boolean	true, false	String	'Gonzo', '', "Gonzo", ""
Integer	42	Double	23.5

Enums

All **Java Enum** types can be used as *primitive* types. To display them, the `toString` method is used. This can be overridden for an *Enum*.

Null

Processing NULL usually leads to errors. There are three ways to avoid these errors in **FRESH MARKER**. The *Default Operator*, the *Existence Operator* and the comparison with the null literal.

The default operator (!) replaces a null value with its optional parameter or the empty string.

The *Existence Operator* (??) returns false, if the value is null, otherwise true.

A comparison with the null literal can be used to check whether a value is null.

Sequences

Everything that implements `List` interface can be used as a *sequence*. Literal Sequences can be used in the template

```
[1, 2, 3, 4, 5, 6, 7]
['eins', 'zwei', 'drei']
```

Hashes

Everything that implements the `Map` interface with `String` keys and also *Beans* and *Records* can be used as *Hash*.

Ranges

A *range* is an numeric interval with a lower limit.

Right-unlimited Range

A *right-unlimited range* has no upper limit. It has the form `x..` where `x` can be any numerical expression.

-10..	lower bound -10
a..	lower bound in the numeric variable <i>a</i>

Right-limited Range

A *right-limited range* has a lower and an upper limit. It has the form `x..y` or `x.<y` where `x` and `y` can be any numeric expression. The second form describes a range with an exclusive upper limit.

-10..<0	lower bound -10 and upper bound -1
a..b	lower bound in variable <i>a</i> and upper bound in <i>b</i>
10..0	lower bound 1 and upper bound 10
a..<b	lower bound in variable <i>a</i> and an exclusive upper bound in <i>b</i>

Length-limited Range

A *length-limited range* is a right-limited range with a different syntax `a..*c`. Here *a* again corresponds to the lower limit of the range and *c* indicates the size of the range.

4..*-5	lower bound 4 and upper bound 0
a..*b	lower bound in variable <i>a</i> and an range size in <i>c</i>

Slices

Slices define slice operators on other data types. These operators can be applied to *Range*, *List*, and *String* values. The *Slices* are described by range expressions in square brackets. The range specifies the lower and upper limits of the *Slice*. With inverted ranges, the slice operation produces an inverted result.

Optionals

Optionals of type `java.util.Optional` are also supported in the model. The empty optionals are interpreted as NULL and all other values are interpreted as instances of generic type.

Lazys

Not all values in the model are used and if their provision is costly, then they can be loaded lazily. **FRESH MARKER** offers the `TemplateObjectSupplier` class for this purpose.

```
Map<String, Object> model = Map.of(
    "tree", TemplateObjectSupplier.of(() -> fromDB(name))
);
template.process(model);
```

Comments

Comments can be placed anywhere inside the template. They can be used to clarify the details of the template.

```
<!-- Comment -->
<#-- Comment -->
```

Expression

Expressions can be used in *Interpolations* and in many other places. A simplified overview shows the possibilities.

```
Expression : OrExpr;
OrExpr : AndExpr ( ('|' | '||' | '^') AndExpr );
AndExpr : EqualityExpr ( ('&' | '&&') EqualityExpr );
EqualityExpr : RelExpr [ ('=' | '==' | '!=' | '<' | '>') RelExpr ];
RelExpr : RangeExpr [ ('>' | '>=' | '<' | '<=') RangeExpr ];
RangeExpr : AddExpr [ ('..' | '..<' | '..*') [ AddExpr ] ];
AddExpr : MultExpr ( ('+' | '-') MultExpr );
MultExpr : UnaryExpr ( ('*' | '/' | '%') UnaryExpr );
UnaryExpr : PlusMinusExpr | NotExpr | DefaultExpr;
PlusMinusExpr : ('+' | '-') DefaultExpr;
NotExpr : "!" DefaultExpr;
DefaultExpr : PrimaryExpr '!' [ PrimaryExpr ];
PrimaryExpr : BaseExpr ( DotKey | DynamicKey
    | MethodInvoke | BuiltIn | Exists );
BaseExpr : Identifier | Literal | Parenthesis
    | BuiltInVar;
Parenthesis : '(' Expression ')';
DotKey : '.' Identifier;
DynamicKey : '[' Expression ']';
MethodInvoke : '(' [ ArgsList ] ')';
BuiltInVariable : '.' Identifier!
```

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Directives

Directives give the template structure. Depending on conditions, their contents are output once, multiple times, or not at all.

List Directive

A *List Directive* prints its contents for each element of a *Sequence* or *Hash*. The optional *looper* variable gives access to the loop metadata. The list can be reduced by the optional *attributes filter*, *offset* and *limit*.

```
<#list sequence as item with looper>
${looper}. ${item}
</#list>
```

The hash keys can be sorted by the optional *sorted* attribute.

```
<#list hash as key sorted asc, values with looper>
${looper}. ${key} $value}
</#list>
```

If Directive

An *If Directive* prints the content for which the first expression in the *If* or one of the optional *Elseif* parts results in true. If no expression applies and there is an optional *Else* part, then its content is printed.

```
<#if number <= 0>zero or less!
<#elseif number == 1>one!
<#else>two or biggerQ
</#if>
```

Switch Directive

A *Switch Directive* prints the content for which the expression in the optional *Case* parts first match the expression in the *Switch* part. If no expression applies and there is an optional *Default* part, then its content is printed.

```
<#switch number>
<#case 0>zero
<#case 1>one
<#default>two or bigger
</#case>
```

A second variant uses *On* parts instead of *Case* parts. These parts can match multiple expressions.

```
<#switch number>
<#on 0, 1>zero or one
<#on 2, 3>two or three
<#default>four or bigger
</#case>
```

...and much more

Further directives (*Brick*, *Setting*, *Outputformat*, *Include*, *Import*, *Var*, *Set*, *Macro*) can be found in the documentation.

Interpolation

Interpolations produce output for the values in the model. The syntax of an interpolation is `${expression}`, with an *Expression* described previously. An *Interpolation* must evaluate to a **FRESH MARKER** *Primitive* type. For example, if the result is a *Sequence*, an exception is thrown. If the result of the evaluation is the special value `NULL`, then an exception is also thrown. An interpolation must always return a result.

Built-Ins

Built-Ins are interpolation operations, which are called on the current value of the interpolation.

The following list is not complete and only shows the more popular built-ins.

Boolean Built-Ins

name	command	output
computer	true?c	true
human	false?h	falsch
then	false?then(23, 42)	42

String Built-Ins

name	command	output
upper case	'test'?upper_case	TEST
lower case	'test'?lower_case	test
capitalize	'test'?capitalize	Test
camel case	'camel-case'?camel_case	camelCase
snake case	'snakeCase'?snake_case	snake_case
kebab case	'kebabCase'?kebab_case	kebab-case
slugify	'One two three'?slugify	one-two-three
trim	' trim '?trim	trim
contains	'test'?contains('t')	true
starts with	'Test'?starts_with('t')	false
ends with	'Test'?ends_with('t')	true
length	'test'?length	4
esc	'<>'?esc('HTML')	<>
left padding	'test'?left_pad(6, '#')	##test
right padding	'test'?right_pad(6, '#')	test##
center padding	'test'?center_pad(6, '#')	##test##
mask	'one two'?mask(2)	*** *to
mask full	'one two'?mask_full('??')	??????

Number Built-Ins

name	command	output
computer	6?c	6
human	6?h	six
format	2.5???format("%.2f")	2.50
absolute	-5?abs	5
sign	-5?sign	-1
min	23?min(42)	23
max	23?min(42)	42
roman	2012?roman	MMXII

Looper Built-Ins

These are *Built-Ins* on the *Looper* Variable in a *List-Directive*. The values change with every loop.

name	command	output
is first	looper?is_first	true
is last	looper?is_last	false
has next	looper?has_next	false
index	looper?index	0
item parity	looper?item_parity	even
item cycle	looper?item_cycle('A', 'B', 'C')	A

Temporal Built-Ins

Number Built-Ins

name	command	output
computer	dateTime?c	1968-08-24T12:34:56
human	yesterday?h	yeasterday
since	yesterday?since	P1D
date	dateTime?date	1968-08-24
time	dateTime?time	12:34:56
string	date?string('dd. MMM')	24. August

...and much more

Further *Built-Ins* for (*Enum*, *Sequence*, *Locale*, *Range*, *Duration*, *Period*, *Version*) can be found in the documentation.

Built-In Variables

Some information is available as *Built-In Variables*. These include the following.

name	description
.now	the current date and time
.locale	the current locale
.version	the current FreshMarker version

Extensions

FreshMarker File File and Path typ support.

FreshMarker Money Money typ support based on Moneta
<https://javamoney.github.io/ri.html>.

FreshMarker Random Random typ support.

Getting started with **FRESH MARKER**

Project: <https://gitlab.com/schegge/freshmarker>

Documentation: <https://schegge.gitlab.io/freshmarker>

Based on a **Overleaf** cheat sheet template <https://overleaf.com>.